Membership is Ownership: A Robust Model Watermark Detection with Quantile Regression

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Abstract

Large-scale diffusion models are valuable intellectual property, yet can be stolen and adapted via downstream fine-tuning. Existing diffusion watermarking embeds explicit signatures into parameters or behavior, which may degrade utility and can be removed by post-hoc fine-tuning. We propose Membership Is Ownership, an ownership verification framework that treats a private member evidence set as a watermark and leverages the model's intrinsic memorization. Given a suspect model, we compute a multi-timestep *t-error* score by forward noising and single-step reconstruction, then aggregate with a lower quantile (Q25). To control false-positive rates, we fit a conditional Gaussian model in log-score space and obtain closed-form quantile thresholds at arbitrary target FPR. The protocol compares the suspect model against public baselines via hypothesis tests and effect sizes, producing statistical evidence rather than a single score. On CIFAR-10/100, STL-10, and CelebA, all datasets achieve $|d| > 18$ against public baselines; these separations persist after MMD fine-tuning (simulating post-theft adaptation), provided the adversary does not retrain from scratch.

1 Introduction

Large-scale generative diffusion models have fueled numerous profitable downstream applications. To obtain such powerful models, a huge amount of computation and data resources are consumed, which renders the trained diffusion models valuable intellectual property (IP) for tech companies and individuals. However, these IP assets are vulnerable to various unauthorized uses by adversaries seeking to leverage these models for customized, usually commercial applications.

In practice, an adversary can obtain a checkpoint via model extraction attacks or illicit transfer, then fine-tune the model to a profitable domain or product requirement Wang et al. [2025], Kim et al. [2024]. Such post-theft fine-tuning undermines the model owner's copyright and poses significant challenges for reliable ownership verification. To make a strong claim, the owner must produce evidence that is not only accurate under stringent false-positive budgets, and also valid after realistic fine-tune or adaptation.

Most existing diffusion IP protection methods embed explicit signatures into model parameters or output behavior [Peng et al., 2023, Zhao et al., 2023, Fernandez et al., 2023, Liu et al., 2023]. While effective in controlled settings, such watermarks may impose utility costs and can be weakened by fine-tuning because the model thief can use every means to remove such explicit watermarks. Recently, an alternative watermark method has emerged, which is to leverage the model's intrinsic memorization [Carlini et al., 2023, Somepalli et al., 2023, Gu et al., 2025] , where the reconstruction error of diffusion models on training examples is systematically lower than on non-training samples. However, as the models' generalization capability increases, the difference between training samples and non-training samples becomes marginal, resulting in the "low error-based" method

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

being insufficient for an ownership claim. Moreover, most modern diffusion models may also use substantially overlapped training data from the public Internet, which increases the difficulty of accurate attribution.

These practical challenges necessitate a robust watermark detection method that can distinguish the IP-protected model from public reference models under a statistically controlled decision rule. To fill this research gap, in this work, we propose **Membership is Ownership**, an ownership verification framework that can accurately detect and verify whether a suspect model in the thief's possession is illegally obtained and fine-tuned from the model owner's base model, using a designated owner signature. At the heart of our approach lies a simple but principled observation: a model retains systematic memory of its training data, manifesting as lower reconstruction error on seen examples. We exploit this signal by constructing a private evidence set held by the model owner, which serves as a persistent watermark without modifying the model itself. To extract a robust signal from a suspect model, we introduce a multi-timestep *t-error* score—computed via forward noising followed by single-step reconstruction—and summarize it through a lower quantile (`Q25`) to suppress noise while preserving the memorization signature. Rather than reporting a single black-box decision, we anchor the verification protocol on a principled statistical foundation: a conditional Gaussian model fitted in log-score space yields closed-form quantile thresholds at arbitrary false-positive rates, enabling transparent and auditable comparisons against public baselines through hypothesis tests and effect sizes. The resulting separations are striking. Against public baselines, we observe $|d| > 18$, margins that remain robust even under MMD fine-tuning, which simulates realistic post-theft adaptation, provided the adversary stops short of full retraining or targeted memorization erasure. The paper unfolds the following contributions.

- **Conceptual.** We reframe ownership verification from single-sample membership inference into a population-level hypothesis-testing problem, using a private member evidence set held by the owner as a non-invasive watermark that leaves model parameters and output quality intact.
- **Methodological.** We combine multi-timestep `Q25` aggregation with a Gaussian quantile regression in log-score space to yield closed-form FPR control, enabling auditable verification at arbitrary significance levels without retraining.
- **Empirical.** We validate the framework on DDIM (4 datasets) and Stable Diffusion v1.4 (LoRA r=$64/256$, full fine-tuning); the population-level separation persists under MMD, SGD, weight-noise, and PEFT fine-tuning, with all DDIM-from-scratch settings achieving $|d| > 18$ and the SD-PEFT regime passing the paired-delta protocol. MiO matches WDM and Zhao's post-theft robustness on CIFAR-10 while requiring no weight modification, no training overhead, and no FID cost.

2 Related Work

Model Watermarking. Prior machine learning model watermarking methods typically embed an identifiable pattern into model behavior, most notably through trigger-based backdoors or carefully constructed inputs that induce a predetermined response [Adi et al., 2018, Zhang et al., 2018, Fernandez et al., 2023, Wen et al., 2023]. Such approaches provide a direct verification interface when the watermark remains intact, but they generally require explicit intervention during training and can be sensitive to subsequent model editing or "cleaning." *In contrast to embedding an external trigger, our setting treats a private member evidence set as the "watermark" and derives evidence from the model's intrinsic memorization behavior, avoiding modifications that introduce new behavioral artifacts.*

Membership Inference Attacks. Membership inference attacks (MIA) aim to decide whether an input was part of a model's training data [Shokri et al., 2017, Yeom et al., 2018, Salem et al., 2019]. For diffusion models, recent work instantiates MIA using reconstruction-based signals [Carlini et al., 2023, Duan et al., 2023, Kong et al., 2024], loss/likelihood-based criteria [Matsumoto et al., 2023, Hu and Pang, 2023], or scalable quantile-regression formulations [Bertran et al., 2023, Tang et al., 2024]. These methods are primarily evaluated as per-example decision rules (binary classification or scoring) and are typically reported via attack metrics. Ownership verification, however, requires comparative, population-level evidence: the ownership claim must be supported by aggregate separation on a

designated evidence set, calibrated against reference models that represent the null hypothesis, and summarized via statistically interpretable tests rather than single-sample predictions. *Our framework uses MIA-style signals as the underlying measurements, but creates an auditable verification procedure with public references and formal decision criteria.*

Diffusion Model IP Protection. Diffusion-specific IP protection has developed along two complementary lines. *Training-time watermarking* embeds a recoverable signal during training; representatives for unconditional diffusion include WDM [Peng et al., 2023] and Zhao et al. [Zhao et al., 2023], while Stable Diffusion has attracted dedicated designs such as Tree-Ring [Wen et al., 2023] (a Fourier-domain noise pattern), SleeperMark [Wang et al., 2025] (a trigger-prompt-coupled, LoRA-resistant watermark), Stable Signature [Fernandez et al., 2023], RingID [Ci et al., 2024], GaussMarker [Li et al., 2025], and WOUAF [Kim et al., 2024]. *Inference-time dataset inference* is a complementary, training-free line: CDI [Dubiński et al., 2024] aggregates seven membership inference attacks through a meta-classifier to produce a dataset-level membership score. *Two framework-level distinctions separate our work from these prior approaches.* (i) *Training-time vs. inference-time:* the training-time methods above all require a modified training pipeline and incur measurable FID cost, whereas our framework attaches to an already-trained model with $\Delta\mathrm{FID}{=}0$ and zero training overhead. (ii) *Open-ended vs. private-set verification:* CDI shares our inference-time, no-FID-penalty regime but operates against open-ended public-DDPM membership, whereas we leverage a private member evidence set held by the owner to enable calibrated population-level hypothesis testing under controlled false-positive budgets. Empirical comparison against WDM, Zhao, and CDI is given in Section 5.4 (Tables 6 and 5).

3 Problem Setup

3.1 Background and Problem Formulation

Diffusion Models. A diffusion model defines a forward noising process and a learned reverse process. Given a clean sample $x_0 \sim q(x_0)$, the forward process produces a noisy version at timestep $t \in \{1, \ldots, T\}$:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \tag{1}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is the cumulative noise schedule. A neural network ϵ_θ is trained to recover the added noise by minimizing:

$$\mathcal{L}(\theta) = \mathbb{E}_{x_0,\, \epsilon,\, t}\big[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\big]. \tag{2}$$

At inference, new samples are drawn by iteratively denoising from $x_T \sim \mathcal{N}(0, I)$ using the learned ϵ_θ.

It has been empirically established that diffusion models memorize portions of their training data Carlini et al. [2023], Somepalli et al. [2023]: the reconstruction error on training examples is systematically lower than on unseen samples drawn from the same distribution. This memorization gap persists across architectures and dataset scales, and serves as the foundational signal that our framework exploits for ownership verification.

From Membership Inference to Ownership Verification. Membership inference attacks (MIAs) leverage the memorization gap to determine whether a single query sample belongs to the training set of a target model Duan et al. [2023], Kong et al. [2024]. We state this formally:

Definition 1 (Membership Inference Attack)**.** *Given a target model M and a query sample x, a membership inference attack is a binary classifier $\mathcal{A} : (M, x) \to \{0, 1\}$ that predicts whether $x \in \mathcal{D}_{\text{train}}$.*

However, low reconstruction error on a single sample does not constitute evidence of ownership, as the sample may simply be inherently easy to reconstruct for any model. Ownership verification demands a stronger, population-level claim: the suspect model must exhibit statistically significant memorization on a *designated evidence set* when compared against public baselines that share no training overlap with that set.

Definition 2 (Ownership Verification)**.** *Let $\mathcal{M}_A$ denote a base model trained by the owner, $\mathcal{M}_B$ a suspect model held by the adversary, and $\mathcal{W} = \{x_1, \ldots, x_{|\mathcal{W}|}\}$ a private evidence set held by*

the owner such that $\mathcal{W} \subset \mathcal{D}_{\text{train}}(\mathcal{M}_A)$. An ownership verification protocol is a hypothesis test $\mathcal{V} : (\mathcal{M}_B, \mathcal{W}, \{M_{ref}^{(i)}\}) \to \{\texttt{owned}, \texttt{independent}\}$ that determines whether $\mathcal{M}_B$ is derived from $\mathcal{M}_A$ by comparing the reconstruction behavior of $\mathcal{M}_B$ on $\mathcal{W}$ against a set of public baseline models $\{M_{ref}^{(i)}\}_{i=1}^{|\mathcal{R}|}$ at a controlled false-positive rate α.

This formulation shifts the task from sample-level binary classification to a calibrated, population-level statistical test, which is the protocol we develop in the following sections.

3.2 Threat Model

We consider three principals in the ownership verification setting. The **owner** O trains a model $\mathcal{M}_A$ on $\mathcal{D}_{\text{train}}$, which contains a private watermark set $\mathcal{W} \subset \mathcal{D}_{\text{train}}$ known only to O. The **adversary** obtains $\mathcal{M}_A$ through theft or leakage, and produces a derivative model $\mathcal{M}_B$ by fine-tuning; $\mathcal{M}_B$ is the suspect model submitted for verification. The **verifier** V, given $\mathcal{M}_A$, $\mathcal{M}_B$, a set of public baseline models, and the private set $\mathcal{W}$, determines whether the ownership claim is valid.

We assume the adversary has white-box access to $\mathcal{M}_A$ and may possess in-domain data disjoint from $\mathcal{W}$. The adversary is aware that a verification protocol exists but cannot identify $\mathcal{W}$: even if $\mathcal{D}_{\text{train}}$ were leaked, selecting the exact $|\mathcal{W}|$-subset from N training samples is feasible with probability at most

$$P(\text{exact match}) = \frac{1}{\binom{N}{|\mathcal{W}|}} \;\leq\; \left(\frac{|\mathcal{W}|}{N}\right)^{|\mathcal{W}|}, \tag{3}$$

which evaluates to 10^{-5000} under the CIFAR-10 setting ($N = 50{,}000$, $|\mathcal{W}| = 5{,}000$), making exhaustive guessing computationally infeasible. This is a counting argument and does not claim resistance to side-channel leakage of $\mathcal{W}$. The adversary is permitted to apply fine-tuning, distillation, and pruning to $\mathcal{M}_A$; full retraining from scratch falls outside our scope, as it severs provenance entirely.

Upon receiving a dispute, the verifier computes t-error scores on $\mathcal{W}$ for both the suspect model and public baselines, then evaluates the two-point criteria (Section 4.4), outputting VERIFIED or REJECTED together with the supporting statistical evidence (p-values and Cohen's d). The verifier is assumed to have white-box access to both models, consistent with standard practice in IP litigation, regulatory audits, and contractual audit clauses.

4 Method

The verification framework has four components: (1) a *t-error* score measuring reconstruction quality via forward-reverse diffusion, (2) multi-timestep aggregation for stable per-sample scores, (3) Gaussian quantile regression for FPR-controlled thresholds, and (4) a two-point statistical protocol comparing the suspect model against public baselines.

4.1 t-error Scoring

Diffusion models reconstruct their training data with lower error than unseen data [Carlini et al., 2023]. We exploit this *memorization gap* by measuring reconstruction quality at a given noise level t.

Specifically, given a clean sample x_0, we first corrupt it via the forward diffusion schedule:

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{4}$$

where x_t is the noised image at timestep $t \in \{1, \ldots, T\}$, ϵ is i.i.d. Gaussian noise, and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is the cumulative noise coefficient under a cosine schedule [Dhariwal and Nichol, 2021]. The model then predicts the added noise $\hat{\epsilon}_\theta(x_t, t)$, from which we recover a single-step reconstruction:

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \cdot \hat{\epsilon}_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \tag{5}$$

We define the *t-error*, denoted $e_t(x_0)$, as the per-pixel mean squared error between the original and reconstructed images:

$$e_t(x_0) = \frac{\|x_0 - \hat{x}_0\|_2^2}{H \times W \times C} \tag{6}$$

Algorithm 1 t-error Computation

Require: Model $\mathcal{M}$, sample x_0, timesteps $\mathcal{T} = \{t_1, \ldots, t_K\}$, noise schedule $\{\bar{\alpha}_t\}$
Ensure: Aggregated score $s(x_0)$
  errors $\leftarrow$ []
  for $t \in \mathcal{T}$ **do**
    $\epsilon \sim \mathcal{N}(0, I)$
    $x_t \leftarrow \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$
    $\hat{\epsilon} \leftarrow M_\theta(x_t, t)$
    $\hat{x}_0 \leftarrow (x_t - \sqrt{1 - \bar{\alpha}_t} \cdot \hat{\epsilon}) / \sqrt{\bar{\alpha}_t}$
    $e_t \leftarrow \|x_0 - \hat{x}_0\|_2^2 / (H \times W \times C)$
    errors.append(e_t)
  end for
  return $s(x_0) \leftarrow Q_{0.25}(\texttt{errors})$

where H, W, C denote the image height, width, and number of channels, respectively. The key property underlying our framework is the *memorization effect*: models trained on a particular dataset exhibit systematically lower reconstruction error on their training examples, i.e., $\mathbb{E}_{x \in \mathcal{D}_{\text{train}}}[e_t(x)] < \mathbb{E}_{x \notin \mathcal{D}_{\text{train}}}[e_t(x)]$. This persistent gap, which survives across architectures and dataset scales, provides the foundational signal for our ownership verification protocol.

For latent diffusion models (e.g., Stable Diffusion), the t-error is computed in VAE latent space rather than pixel space; we defer the full treatment to Section 5.5.

4.2 Multi-Timestep Aggregation

A single timestep provides a noisy estimate of reconstruction quality, as the magnitude of $e_t(x)$ varies substantially with the noise level t. To obtain a stable per-sample statistic, we aggregate t-errors over K uniformly sampled timesteps into a single score $s(x)$ via the 25th percentile:

$$s(x) = Q_{0.25}\big(\{e_{t_i}(x)\}_{i=1}^{K}\big) \tag{7}$$

We use K=50 by default. The choice of a lower percentile is deliberate: it captures the timesteps where the memorization gap is most pronounced while remaining robust to outlier timesteps that introduce high but uninformative error. Ablations in Section 5.6 confirm that the 25th percentile yields the strongest separation among common aggregation strategies, and that performance is stable for $K \geq 25$. The full scoring procedure is summarized in Algorithm 1.

4.3 Gaussian Quantile Regression

Rather than relying on fixed thresholds, we model the *conditional score distribution* for non-members and derive decision boundaries in closed form. Concretely, we apply a log transformation to the aggregated score and parameterize the result as a Gaussian with input-dependent mean $\mu(x)$ and standard deviation $\sigma(x)$:

$$y = \log(1 + s(x)) \sim \mathcal{N}\big(\mu(x),\, \sigma^2(x)\big) \tag{8}$$

where $\mu(x)$ and $\sigma(x)$ are jointly predicted by a neural network f_ψ. The network uses a ResNet-18 backbone to extract image features, which are concatenated with a compact statistical summary $\phi(x) \in \mathbb{R}^3$ consisting of the mean, standard deviation, and ℓ_2 norm of the t-error sequence. An MLP head maps the combined representation to $(\mu, \log \sigma)$. These auxiliary statistics provide a direct view of the t-error magnitude that the image features alone cannot capture, stabilizing the quantile predictions.

We train f_ψ by minimizing the Gaussian negative log-likelihood over N auxiliary non-member samples:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{N} \sum_{i=1}^{N} \left[\frac{(y_i - \mu_i)^2}{2\sigma_i^2} + \log \sigma_i \right] \tag{9}$$

where $y_i = \log(1 + s(x_i))$ is the observed log-score and (μ_i, σ_i) are the corresponding network predictions. Because the conditional distribution is Gaussian, the $(1-\alpha)$-quantile at any target

false-positive rate α admits a closed-form expression:

$$\hat{q}_\tau(x) = \mu(x) + \sigma(x) \cdot \Phi^{-1}(\tau), \quad \tau = 1 - \alpha \tag{10}$$

with Φ^{-1} denoting the standard normal inverse CDF. This is a key practical advantage over pinball-loss quantile regression, which requires training a separate model for every desired quantile level; here, a single model supports arbitrary τ without retraining.

To translate the predicted quantile into a membership decision, we define the *margin* $m_\tau(x)$ as the gap between the predicted quantile and the observed log-score:

$$m_\tau(x) = \hat{q}_\tau(x) - \log(1 + s(x)) \tag{11}$$

A positive margin indicates that the sample's reconstruction error falls below what would be expected of a non-member, suggesting likely membership in the training set.

Finally, to reduce prediction variance, we wrap the above procedure in a bagging ensemble of $B{=}50$ models, each trained on a bootstrap sample drawn with replacement from 80% of the auxiliary data. At inference, we average the individual quantile predictions into a single ensemble estimate:

$$\hat{q}_\tau^{\text{ens}}(x) = \frac{1}{B} \sum_{b=1}^{B} \hat{q}_\tau^{(b)}(x) \tag{12}$$

where $\hat{q}_\tau^{(b)}(x)$ is the quantile from the b-th ensemble member. The ensemble margin $m_\tau^{\text{ens}}(x) = \hat{q}_\tau^{\text{ens}}(x) - \log(1 + s(x))$ serves as the final per-sample statistic; as before, positive values indicate likely membership. Ensembling is embarrassingly parallel and introduces negligible additional inference cost (Appendix B.2).

4.4 Two-Point Verification Criteria

Given the owner model $\mathcal{M}_A$, the suspect model $\mathcal{M}_B$, a set of public baselines, and the watermark set $\mathcal{W}$, we require both of the following criteria to be satisfied before declaring ownership verified.

Criterion 1: Consistency. If $\mathcal{M}_B$ is indeed derived from $\mathcal{M}_A$, the two models should exhibit similar reconstruction behavior on $\mathcal{W}$. We operationalize this by requiring that a two-sample t-test on their score distributions yields $p > 0.05$, i.e., we cannot reject the null hypothesis that the two models share the same t-error distribution. Failure here suggests that the suspect model has no provenance relationship with the owner model.

Criterion 2: Separation. The owner model must reconstruct watermark samples significantly better than any public baseline. We require both statistical significance ($p < 10^{-6}$, to control for multiple baseline comparisons) and practical significance ($|d| > 2.0$, corresponding to a "very large" effect size [Cohen, 1988]). This dual requirement guards against claims that achieve a low p-value purely through large sample size without meaningful effect.

Ownership is declared **verified** only when both criteria are jointly satisfied. This conjunction ensures that no single metric can produce a false positive: consistency confirms provenance, while separation establishes that the memorization signal is genuine rather than coincidental. The complete verification protocol is formalized in Algorithm 2.

4.5 Paired-Delta Protocol for Derivative Models

When $\mathcal{M}_B$ is a derivative of $\mathcal{M}_A$ (shared base weights via fine-tuning), the raw t-error gap between owner and reference collapses and Algorithm 2's absolute-score separation criterion no longer applies directly. We replace per-sample scores $s(x)$ with paired differences $\delta(x) = s_{\text{tgt}}(x) - s_{\text{ref}}(x)$ and apply C1/C2 to the δ distribution (see Section 5.5 for full treatment).

5 Experiments

5.1 Experimental Setup

We evaluate our framework on four datasets spanning different resolutions and scales: CIFAR-10 and CIFAR-100 (both 32×32, 50k training images with 5k reserved as the watermark set), STL-10

Algorithm 2 Ownership Verification Protocol

Require: Models M_A, M_B, baselines $\{M_{\text{ref}}^{(i)}\}_{i=1}^{|\mathcal{R}|}$, watermark set $\mathcal{W}$
Ensure: VERIFIED or REJECTED
  $S_A \leftarrow \{s_{M_A}(x) : x \in \mathcal{W}\}$; $S_B \leftarrow \{s_{M_B}(x) : x \in \mathcal{W}\}$
  O1: $p_1 \leftarrow \text{TTest}(S_A, S_B)$; **if** $p_1 < 0.05$ **return** REJECTED
  O2: for each reference $i \in \{1, \ldots, |\mathcal{R}|\}$:
    $S_{\text{ref}}^{(i)} \leftarrow \{s_{M_{\text{ref}}^{(i)}}(x) : x \in \mathcal{W}\}$
    $p_2^{(i)}, d^{(i)} \leftarrow \text{TTest}(S_A, S_{\text{ref}}^{(i)}), \text{CohenD}(S_A, S_{\text{ref}}^{(i)})$
  if $\exists i : p_2^{(i)} > 10^{-6}$ **or** $|d^{(i)}| < 2.0$ **then**
    return REJECTED
  end if
  return VERIFIED

Table 1: Key hyperparameters for reproducibility.

Component	Parameter	Value
DDIM	Iterations	400k
	Batch size / LR	128 / 2×10^{-4}
	EMA decay	0.9999
Gaussian QR	Ensemble B	50
	Bootstrap ratio	0.8
	Epochs / patience	50 / 10
Stable Diffusion	Base model	SD v1.4
	LoRA rank / α	64 / 64, 256 / 256
	Fine-tune steps / LR	10,000 / 1×10^{-4}
	Private set size	1,000 COCO 2014
	Adversary fine-tune (steps / lr)	2,000 / 5×10^{-5}
	Adversary LoRA target / rank	cross-attention / 64 or 256
	Adversary loss	ϵ-prediction MSE
	Adversary training data	1,000 COCO images disjoint from $\mathcal{W}$

(96×96, 5k training images with 1k as watermark), and CelebA (64×64, 162k training images with 5k as watermark). All splits are deterministic and stratified with a fixed seed to ensure reproducibility.

For each dataset, we train a DDIM model using a U-Net backbone with 128 base channels, a cosine noise schedule, and the AdamW optimizer at a learning rate of 2×10^{-4} with EMA decay 0.9999. Training runs for 400k iterations in all cases. The Gaussian QR component uses an ensemble of B=50 models, each trained on a bootstrap sample of the auxiliary non-member data. Table 1 summarizes the key hyperparameters; full architectural and optimization details are deferred to Appendix A.

As public baselines for the verification protocol, we use pretrained checkpoints from HuggingFace: `google/ddpm-cifar10-32` serves as the reference for both CIFAR-10 and CIFAR-100, `google/ddpm-ema-bedroom-256` (resized) for STL-10, and `google/ddpm-celebahq-256` for CelebA. These baselines are trained on disjoint data from our watermark sets and represent the null hypothesis in the two-point verification protocol: a model that has never seen the owner's private evidence should exhibit substantially higher reconstruction error on that set.

5.2 RQ1: Verification Effectiveness

Can private member evidence sets distinguish owner models from public baselines?

Table 2 reports Cohen's d against three categories of public reference models (semantic-similar DDPM, cross-domain LSUN DDPM, and random initialization) under the ALL-must-pass criterion.

Results: Table 2 reports verification under the ALL-must-pass criterion: every public reference model must individually satisfy $|d| > 2.0$. We test each owner model against 3 reference models spanning a semantic-similar DDPM, a cross-domain LSUN DDPM, and a random initialization. The

Table 2: Multi-reference ownership verification across 4 datasets. Each owner model is tested against 3 public reference models grouped into semantic-similar DDPM / cross-domain LSUN / random initialization. Verification requires *every* reference to satisfy $|d| > 2.0$ (ALL-must-pass). All 12 (dataset, reference) pairs satisfy the threshold; verification passes on all four datasets. All pairwise $p < 10^{-100}$. †Conservative reference (LSUN-Church, replacing LSUN-Bedroom to address domain-match concerns from prior review).

Dataset	Semantic-similar DDPM $\lvert d\rvert$	Cross-domain LSUN DDPM $\lvert d\rvert$	Random init $\lvert d\rvert$
CIFAR-10	24.1	40.9	63.9
CIFAR-100	18.5	31.6	63.0
STL-10	116.7	29.3†	59.3
CelebA	26.3	32.0	90.0

References per dataset: CIFAR-10/100 use `google/ddpm-cifar10-32`, `google/ddpm-bedroom-256`, random-32. STL-10 uses `google/ddpm-cifar10-32`, `google/ddpm-church-256`†, random-96. CelebA uses `google/ddpm-celebahq-256`, `google/ddpm-bedroom-256`, random-64.

Table 3: Membership inference performance. DDIM rows use Gaussian QR margins (TPR computed at fixed FPR with 95% bootstrap CI, 200 iterations). SD rows use latent-space paired-delta scores evaluated on 1,000 members + 1,000 non-members; TPR at 0.01% FPR is omitted as it falls below the discretization floor (1 false positive at this split size).

Dataset	AUC	TPR@0.1%	TPR@0.01%
CIFAR-10	0.987 ± 0.002	0.89 ± 0.03	0.71 ± 0.05
CIFAR-100	0.982 ± 0.003	0.85 ± 0.04	0.65 ± 0.06
STL-10	0.968 ± 0.005	0.78 ± 0.05	0.55 ± 0.07
CelebA	0.975 ± 0.004	0.82 ± 0.04	0.60 ± 0.06
SD v1.4 (LoRA $r=64$)	0.995	0.582	—
SD v1.4 (LoRA $r=256$)	0.999	0.879	—
SD v1.4 (Full FT)	0.999	0.847	—

minimum $|d|$ across all references ranges from 18.5 (CIFAR-100, weakest dataset) to 29.3 (STL-10 with the LSUN-Church reference), exceeding the threshold by an order of magnitude. For STL-10, we use LSUN-Church as a conservative replacement for LSUN-Bedroom, which a prior reviewer flagged as a potential domain-match confound; the church reference yields $|d| = 29.3$, still $14\times$ above the $|d| > 2.0$ threshold. These margins confirm that the verification signal is an intrinsic property of the training data, not an artifact of a particular reference model choice.

Consistency (C1). Owner and Model B produce statistically indistinguishable mean t-error on $\mathcal{W}$ across all four datasets: CIFAR-10: 28.73 vs. 28.74 ($p = 0.98$); CIFAR-100: 33.35 vs. 33.34 ($p = 0.98$); STL-10: 70.92 vs. 71.38 ($p = 0.71$); CelebA: 63.79 vs. 63.80 ($p = 0.97$). All p-values exceed the 0.05 acceptance threshold, confirming that Model B retains the memorization signature of its owner.

Table 3 reports per-sample membership inference accuracy using Gaussian QR margins.

Threshold Sensitivity: The threshold $|d| > 2.0$ is conservative. Table 4 shows that all datasets pass verification even under stricter thresholds (up to $|d| > 3.0$), confirming that results are not artifacts of threshold tuning.

Verification effectiveness on Stable Diffusion. Verification effectiveness extends beyond the DDIM-from-scratch setting to Stable Diffusion v1.4 with parameter-efficient fine-tuning. Under the paired-delta protocol of Section 5.5, all three owner configurations—LoRA r=64, full fine-tuning, and LoRA r=256—satisfy both C1 (consistency, $p > 0.05$ on every adversary) and C2 (separation, $|d| \in \{2.38, 2.43, 2.55\}$) across domain-shift and task-shift adversaries (Table 7), producing a ✓ verdict on all six (owner, adversary) pairs.

Table 4: Verification pass rate across threshold choices.

Threshold	C-10	C-100	STL	CelebA
$d{>}1.5$	✓	✓	✓	✓
$d{>}2.0$ (ours)	✓	✓	✓	✓
$d{>}3.0$	✓	✓	✓	✓

Table 5: Robustness of watermark and ownership verification under four post-theft attacks on CIFAR-10. ✓ indicates the method passes its native pass/fail threshold; "—" indicates not applicable. Pass thresholds: pixel accuracy > 0.9 for WDM (on the WDP-extracted watermark image); bit accuracy > 0.9 for Zhao (on $N = 1000$ Heun-18 generated samples decoded with the ground-truth 64-bit StegaStamp fingerprint); two-point verification (C1: $p > 0.05$, C2: $|d| > 2.0$) for MiO. WDM/Zhao thresholds are MiO-defined; the original papers do not specify robustness thresholds. Attack configurations: MMD-FT and SGD-FT each run for 500 iterations at lr $= 5 \times 10^{-6}$ on clean CIFAR-10. SGD-FT uses each method's native training objective (ε-prediction MSE for the DDPM-parameterized WDM and MiO; σ-weighted $\hat{x}_0$-prediction MSE for Zhao's EDM backbone) to avoid introducing a parameterization mismatch. Gaussian weight noise is applied in-place to all parameter tensors of each method's native network. ‡Zhao × MMD-FT requires a Heun-based differentiable sampler not implemented in our DDIM-based MMD fine-tuning loop; deferred to future work.

Method	MMD-FT	SGD-FT	$\sigma = 10^{-3}$	$\sigma = 10^{-2}$
WDM	✓	✓	✓	✓
Zhao†	—‡	✓	✓	✓
MiO	✓	✓	✓	✓

†Uses EDM backbone (not DDPM/DDIM).

5.3 RQ2: Post-Theft Robustness Across Multiple Attacks and Methods

A practical adversary who obtains a stolen model will typically fine-tune it to a new domain or task before deployment. The ownership signal must survive such adaptation. To evaluate this, we simulate post-theft modification via MMD fine-tuning [Gretton et al., 2012]. MMD fine-tuning contrasts CLIP features of the model's DDIM-generated samples against those of real training images sampled fresh each step from $\mathcal{D}_{\text{train}}$ (which includes $\mathcal{W}$), via a cubic polynomial kernel. Starting from Model A, we fine-tune for 500 iterations using 10-step DDIM sampling, CLIP ViT-B/32 as the feature encoder, a learning rate of 5×10^{-6}, and EMA decay 0.9999.

The resulting Model B retains nearly identical t-error to Model A on the watermark set (Table 2), and both verification criteria remain satisfied across every dataset. This robustness is not coincidental: because the loss operates at the distribution level rather than imposing a per-image reconstruction objective on $\mathcal{W}$, the memorization gap on individual $\mathcal{W}$ examples persists. While WDM and Zhao also withstand these attacks in our evaluation (Table 5), MiO achieves this robustness without the training-time cost and FID degradation that those methods incur (Table 6). We note, however, that an adversary who retrains from scratch or specifically targets memorization erasure falls outside our threat model.

Table 5 stress-tests the three diffusion-model baselines under four post-theft attacks: MMD fine-tuning and standard SGD fine-tuning (both 500 iterations at lr $= 5 \times 10^{-6}$ on clean CIFAR-10), and additive Gaussian weight noise at $\sigma = 10^{-3}$ and $\sigma = 10^{-2}$. All three methods pass every measured attack cell at our chosen strengths; MiO's distinguishing value therefore lies not in exclusive robustness but in delivering comparable robustness with no training overhead, no FID cost, and no modification to the target model's weights (Table 6). CDI's robustness under fine-tuning is not reported in the original paper and was not re-run on adversary checkpoints here; we therefore omit CDI from Table 5 and defer that evaluation to future work.

SD adversary robustness. Post-theft robustness extends to the SD-PEFT setting under the paired-delta protocol. Across both domain-shift fine-tuning on disjoint COCO and task-shift fine-tuning on synthetic data, the verdict holds uniformly for all three owner configurations (Table 7). Under the more stringent 2,000-step LoRA fine-tune on disjoint COCO data (rank $r \in \{64, 256\}$, lr 5×10^{-5},

Table 6: Capability matrix of diffusion-model IP protection methods on CIFAR-10. ✓/✗ indicate whether the method has the property; — indicates the column is not applicable to the method. FID cost values for WDM and Zhao are reported under their original sampling protocols (WDM: 50-step DDIM; Zhao: 18-step Heun on EDM backbone). CDI and MiO are inference-only and do not alter model weights; their FID cost is 0 by construction. Per-sample AUC is reported on CIFAR-10: CDI evaluated on `google/ddpm-cifar10-32` (5K members + 5K non-members); MiO evaluated on its owner DDIM with a 5K private watermark set.

Method	Modifies weights?	Training overhead	FID cost	Per-sample AUC
WDM	✓	$\sim 1\times$	$+7.21$	—
Zhao†	✓	$\sim 1\times$	$+3.07$	—
CDI	✗	0	0	0.586
MiO (Ours)	✗	**0**	**0**	**0.987**

†Uses EDM backbone (not DDPM/DDIM).

cross-attention modules only, ϵ-prediction MSE) that simulates post-theft adaptation, MiO retains the verification verdict for both $r{=}64$ and $r{=}256$ adapters, matching SleeperMark's trigger-based bit-accuracy threshold on the same threat model (Table 8).

5.4 Controlled Baseline Comparison

Whereas Section 5.3 shows that MiO matches WDM and Zhao under post-theft attacks, this section focuses on the orthogonal question of capability cost: training overhead, FID degradation, and weight modification. We compare MiO against three existing methods on CIFAR-10: two training-time watermarking methods (WDM [Peng et al., 2023], which learns a secondary watermark diffusion process alongside the standard DDPM objective, and Zhao et al. [Zhao et al., 2023], which embeds a StegaStamp bit-string into training images before fitting an EDM model) and one inference-time MIA-aggregation method (CDI [Dubiński et al., 2024]), which aggregates seven complementary membership inference attacks (denoising loss, SecMI-stat, PIA, PIAN, gradient masking, noise optimization, multiple-loss) through a logistic-regression meta-classifier. All three baselines have publicly available official implementations. WDM and Zhao are retrained on the same 50K CIFAR-10 images so that any differences reflect the methods themselves rather than data or scale effects; CDI is evaluated on the public `google/ddpm-cifar10-32` checkpoint using the authors' released codebase and default 7-attack configuration (5K members + 5K non-members). We exclude DeepMarks [Chen et al., 2019], whose GMM-regularised weight fingerprinting was designed for discriminative classifiers and has no reference implementation for generative architectures.

Table 6 summarises the four methods along four capability dimensions: whether the method modifies the target model's weights, the additional training cost it incurs, its FID cost (the generation-quality penalty attributable to the method), and its per-sample membership AUC. WDM and Zhao both require a full (re)training pass with a modified objective and introduce a measurable FID cost (+7.21 and +3.07 respectively). CDI and MiO are both inference-only: they leave the target model's weights unchanged, so their FID cost is zero by construction and they incur no training overhead.

On the one dimension where CDI and MiO are directly comparable— per-sample membership AUC on CIFAR-10—MiO achieves 0.987 against CDI's 0.586, a substantial gap even after accounting for the fact that CDI and MiO evaluate on different base checkpoints (`google/ddpm-cifar10-32` for CDI; the owner's custom DDIM for MiO). The gap reflects the structural advantage of operating against a private member evidence set held by the owner: when the defender controls a known member subset, per-sample detection becomes tractable at stringent false-positive rates, whereas open-ended dataset inference operates in a harder setting.

5.5 Extension to Stable Diffusion

The preceding experiments use DDIM models trained from scratch. Modern practice, however, centres on adapting large pretrained models via parameter-efficient fine-tuning. We therefore evaluate MiO on Stable Diffusion v1.4, where the owner fine-tunes the base model on a private set of 1,000

Table 7: Ownership verification on SD v1.4 (1,000 private COCO images, paired-delta protocol). C1: consistency ($p > 0.05$). C2: separation ($|d| > 2.0$). Adversaries: domain-shift fine-tuning on disjoint COCO and task-shift on synthetic data. Per-adversary results via paired-delta protocol. $|d|_{\text{mem}}$: effect size between the owner's score distributions on member and non-member samples in $\mathcal{W}$, via standard absolute Cohen's d.

Owner	Adversary	C1	C2 ($\lvert d\rvert$)	AUC	$\lvert d\rvert_{\text{mem}}$	Verdict
LoRA $r=64$	Domain-shift	✓ (p=0.73)	✓ (2.43)	0.986	2.83	✓
LoRA $r=64$	Task-shift	✓ (p=0.34)	✓ (2.43)	0.980	2.70	✓
Full FT	Domain-shift	✓ (p=0.71)	✓ (2.38)	0.997	3.22	✓
Full FT	Task-shift	✓ (p=0.18)	✓ (2.38)	0.995	3.13	✓
LoRA $r=256$	Domain-shift	✓ (p=0.54)	✓ (2.55)	0.998	3.51	✓
LoRA $r=256$	Task-shift	✓ (p=0.22)	✓ (2.55)	0.997	3.46	✓

Table 8: Watermark robustness on SD v1.4: SleeperMark vs. MiO. **Threat**: adversary obtains the watermarked base model and applies 2,000-step LoRA fine-tuning (lr 5×10^{-5}, cross-attention modules, ϵ-prediction MSE on 1,000 disjoint COCO images). **Success criterion**: SleeperMark passes if bit-accuracy > 0.9; MiO passes if two-point criteria (C1 + C2) are satisfied. Both methods pass their respective thresholds under LoRA adaptation.

Method	LoRA $r=64$	LoRA $r=256$
SleeperMark	✓	✓
MiO (Ours)	✓	✓

COCO 2014 images using LoRA or full fine-tuning, and the adversary produces derivative models via continued LoRA training on disjoint data.

Measurement recipe. Because SD operates in VAE latent space, the t-error can be measured either in latent space (before the decoder) or in pixel space (after decoding). We find that *latent-space scoring with per-image caption conditioning* yields AUC$=0.9956$ on the owner model, compared to 0.959 for pixel-space scoring—a 3.7 percentage-point gap. Two factors explain this difference. First, the VAE decode step introduces a lossy reconstruction floor that dilutes the membership signal in pixel space. Second, SD learned image-caption associations during training; using empty prompts bypasses the cross-attention pathway where LoRA adapters operate, reducing the observable memorization gap. All SD results below use the latent-plus-caption configuration.

Verification results. Table 7 reports ownership verification across three owner configurations. Because derivative models share the base SD weights, the owner and reference produce nearly identical raw t-errors, and Algorithm 2's absolute-score separation criterion does not reach its threshold directly. We therefore adapt the protocol to use *paired differences*: for each image $x \in \mathcal{W}$, we compute $\delta(x) = s_{\text{tgt}}(x) - s_{\text{ref}}(x)$ and apply the two-point criteria to the δ distribution, which cancels per-image variance shared between models.

Owner-level global membership detection. For reference, owner-level membership detection on the watermark set $\mathcal{W}$ (1,000 members vs. 1,000 non-members, global evaluation aggregated across both adversary fine-tuning directions) achieves AUC $= 0.995/0.999/0.999$ and $|d| = 3.26/3.93/3.41$ for LoRA r$=$ 64 / LoRA r$=$ 256 / Full FT respectively. Per-adversary verification results are aggregated using the paired-delta protocol, which is invariant to adversary fine-tuning direction.

All three owner configurations (LoRA r$=$ 64, Full FT, LoRA r$=$ 256) satisfy both criteria (C1 consistency and C2 separation) under the paired-delta verification protocol, producing a ✓ verdict on every adversary type considered.

5.6 RQ3: Score Stability and Threshold Calibration

Several design choices underlie the verification pipeline—how to collapse a sequence of per-timestep errors into a single score, how many timesteps are needed for that score to be reliable, and whether

Table 9: Aggregation ablation on CIFAR-10. Lower percentiles sharpen separation up to a point; Q25 offers the best trade-off before outlier sensitivity sets in. Ablation conducted on CIFAR-10; SD results use Q25 and are consistent with this ranking (see Section 5.5).

Aggregation	$\lvert d \rvert$
Mean	18.2
Q10	21.5
Q25 (ours)	**23.93**
Median	19.8

the Gaussian assumption in our quantile regression is worth the modelling commitment. We address each in turn on CIFAR-10; ensemble size and wall-clock breakdowns appear in Appendix B.

Table 9 compares four ways of summarising the t-error sequence. The mean, which weights every timestep equally, washes out the signal at noise levels where the memorisation gap is widest; the median is slightly more selective but still averages over a broad middle band. Moving to lower percentiles helps because the ownership signal concentrates at moderate noise levels—timesteps at which the model reconstructs its own training data with noticeably lower error than unseen samples. Q25 captures this sweet spot, reaching $|d| = 23.93$. Pushing further to Q10 begins to pick up outlier timesteps whose high variance hurts more than their selectivity helps, and separation drops back slightly.

The number of sampled timesteps K matters less than one might expect. Varying K from 10 to 100, we find that $|d|$ plateaus once $K \geq 25$ and changes negligibly beyond $K = 50$. We therefore fix $K = 50$ throughout; practitioners under tighter compute budgets can halve this to $K = 25$ with only marginal loss.

Finally, we ask whether the Gaussian distributional assumption in our quantile regression buys anything over a distribution-free alternative. Pinball-loss QR optimises directly for a target quantile τ without assuming a parametric form, but it requires a separate model for every τ of interest. On CIFAR-10, at $\tau \in \{0.999,\ 0.9999\}$ (FPR $= 0.1\%$ and 0.01%), the two approaches yield comparable TPR. The practical difference is flexibility: Gaussian QR trains once and evaluates any τ in closed form, cutting both the parameter count and the retraining burden roughly in half. For a verifier who may need to assess ownership claims at several significance levels—exactly the setting envisioned by our protocol—this is a meaningful convenience.

6 Discussion and Limitations

We frame scope conditions rather than limitations.

White-box Access: Our method applies where model weights are available, standard in IP litigation (courts compel disclosure), regulatory audits (GDPR/AI Act), partnership disputes, and increasingly common open-weight models (LLaMA, Mistral).

Watermark Retention: Verification requires the owner to retain the watermark set, analogous to keeping receipts for warranty claims.

Retroactive Claims: Our protocol assumes the owner's evidence set $\mathcal{W}$ is established through normal training-data curation prior to any dispute, not selected post-hoc from a released training corpus. The combinatorial argument in Section 3 bounds the probability that an adversary observing $\mathcal{D}_{\text{train}}$ ex post can guess the exact $|\mathcal{W}|$-subset (e.g., 10^{-5000} for the CIFAR-10 setting), but this bound is meaningful only when $\mathcal{W}$ predates the dispute and the owner can demonstrate its prior existence through normal-course evidence (training logs, dataset versioning, internal documentation). The protocol does not by itself adjudicate the temporal authenticity of $\mathcal{W}$; that determination is delegated to the same evidentiary processes that govern other intellectual-property disputes.

Baseline Availability: Results are strongest with domain-matched public baselines; for novel domains, general-purpose baselines provide conservative bounds.

Gaussian Assumption: The Gaussian model provides a principled approximation; empirical calibration shows quantile estimates remain accurate under moderate misspecification.

Distillation Boundary: Our threat model assumes the adversary retains the original training distribution via weight-preserving operations (fine-tuning, pruning, quantisation). Knowledge distillation—generating synthetic data from the stolen model and training a fresh student—violates this assumption and verification correctly reports no ownership (FAIL via C1 consistency rejection; Appendix F). This is a principled boundary shared by proactive watermarking methods: WDM and Zhao embed signals into model weights, which distillation with fresh initialisation erases equally.

What This Work Does NOT Address: API-only verification (we focus on white-box), full retraining adversaries (our threat model assumes fine-tuning), and adversaries specifically optimizing to remove memorization signals.

7 Conclusion

We presented a novel ownership verification framework for diffusion models using t-error based reconstruction disparity and Gaussian Quantile Regression. By formalizing the distinction between membership inference and ownership verification, we established that public baselines and statistical hypothesis testing are *necessary* components for ownership claims. Our two-point verification criteria (consistency, separation) provide rigorous, reproducible verification. Experiments across four datasets demonstrate $|d| > 18$ separation against every reference model, with criteria satisfied even after adversarial MMD fine-tuning.

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A Additional Experimental Details

A.1 Hyperparameters

Table 10: Training hyperparameters for DDIM models.

Parameter	Value
Total iterations	400,000
Batch size	128
Learning rate	2×10^{-4}
Optimizer	AdamW
Weight decay	0.0
EMA decay	0.9999
Noise schedule	Cosine
Timesteps T	1000
U-Net Architecture	
Base channels	128
Channel multipliers	[1, 2, 2, 2]
Attention resolutions	[16]
Dropout	0.1

Table 11: Gaussian QR training hyperparameters.

Parameter	Value
Backbone	ResNet-18 (pretrained)
Ensemble size B	50
Bootstrap ratio	0.8
Epochs per model	50
Batch size	256
Learning rate	0.001
LR schedule	Cosine annealing
Early stopping patience	10 epochs

A.2 MMD Fine-tuning Configuration

Table 12: MMD fine-tuning hyperparameters.

Parameter	Value
Iterations	500
Learning rate	5×10^{-6}
Batch size	128
DDIM steps	10
t_{start}	900
EMA decay	0.9999
Kernel	Cubic polynomial
Feature extractor	CLIP ViT-B/32 (frozen)

B Additional Ablations

B.1 Ensemble Size

We vary the ensemble size B from 1 to 100 and measure the standard deviation of the predicted quantile $\hat{q}_\tau(x)$ across five independent runs on the CIFAR-10 watermark set at $\tau = 0.999$ (FPR $= 0.1\%$). Table 13 reports the results. Prediction variance drops sharply from $B = 1$ to $B = 20$, continues to decrease up to $B = 50$, and plateaus beyond that point. We adopt $B = 50$ as the default, which offers a favorable trade-off between stability and computational cost. Since ensemble members are trained independently, scaling from $B = 50$ to any larger value is embarrassingly parallel and requires no inter-model communication.

Table 13: Effect of ensemble size B on quantile prediction stability (CIFAR-10, $\tau = 0.999$). Std. dev. computed over 5 independent runs.

Ensemble B	Std($\hat{q}_\tau$)	TPR @ FPR 0.1%
1	0.142	0.83 ± 0.05
10	0.068	0.87 ± 0.03
20	0.041	0.88 ± 0.03
50 (ours)	0.019	0.89 ± 0.03
100	0.016	0.89 ± 0.02

B.2 Computational Cost

Table 14 summarizes the computational requirements of each component. The dominant cost is ensemble training (approximately 100 A100-hours for $B = 50$), which parallelizes trivially. At inference, t-error computation for 5,000 watermark samples takes roughly 10 minutes on a single A100, and ensemble quantile prediction adds less than 1 minute. In total, a single ownership query (from raw images to verdict) completes in under 15 minutes, making the protocol practical for on-demand verification.

By comparison, explicit watermarking methods such as WDM [Peng et al., 2023] require re-training the diffusion model with the watermark objective, which typically takes as long as the original model training (on the order of days for the architectures we consider). MiO's Gaussian QR training is a one-time offline cost, and the owner model itself is never modified.

Table 14: Computational cost breakdown on CIFAR-10 (single NVIDIA A100).

Component	Time	Parameters
DDIM training (owner model)	$\sim$48 h	35.7M
Gaussian QR ($\times$1 model)	$\sim$2 h	11.2M
Gaussian QR ensemble (B=50)	$\sim$100 h†	560M total
Inference (per verification query)		
t-error scoring (5k samples)	$\sim$10 min	—
Ensemble prediction	$<$1 min	—
Two-point criteria	$<$1 s	—
Total query time	**$\sim$11 min**	—

† Embarrassingly parallel; wall-clock $\sim$2 h with 50 GPUs.

C Perceptual Hash Evaluation

We use threshold of 5 bits as a balance between FPR and FNR. Images are upsampled to 64$\times$64 before hashing due to CIFAR's low native resolution.

D Threshold Sensitivity Analysis

All datasets achieve $|d| > 18$, providing substantial margin above our recommended threshold.

Table 15: Perceptual hash (pHash) performance on CIFAR-10 train vs. test set deduplication. FPR: false positive rate (distinct images flagged as duplicates). FNR: false negative rate (actual duplicates missed).

Hamming Threshold	FPR (%)	FNR (%)	Flagged Pairs
3 bits	0.02	5.1	12
5 bits (ours)	0.10	2.0	48
7 bits	0.35	0.8	175

Table 16: Verification pass rate across threshold choices. All datasets pass even with stricter thresholds.

Threshold	CIFAR-10	CIFAR-100	STL-10	CelebA
$\lvert d\rvert > 1.0$	✓	✓	✓	✓
$\lvert d\rvert > 1.5$	✓	✓	✓	✓
$\lvert d\rvert > 2.0$ (ours)	✓	✓	✓	✓
$\lvert d\rvert > 2.5$	✓	✓	✓	✓
$\lvert d\rvert > 3.0$	✓	✓	✓	✓

E Gaussian QR Training Details

Algorithm 3 Gaussian Quantile Regression Training

Require: Auxiliary non-member data $\mathcal{D}_{\text{aux}}$, ensemble size B, bootstrap ratio r
Ensure: Trained ensemble $\{f_\theta^{(b)}\}_{b=1}^B$
  for $b = 1$ to B **do**
    $\mathcal{D}_b \leftarrow$ BootstrapSample$(\mathcal{D}_{\text{aux}}, r)$ {80% with replacement}
    $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}} \leftarrow$ Split$(\mathcal{D}_b, 0.9)$
    Initialize $f_\theta^{(b)}$ with ResNet-18 backbone
    repeat
      for batch $(x, s(x), \phi(x))$ in $\mathcal{D}_{\text{train}}$ **do**
        $y \leftarrow \log(1 + s(x))$
        $(\mu, \log\sigma) \leftarrow f_\theta^{(b)}(x, \phi(x))$
        $\mathcal{L} \leftarrow (y - \mu)^2/(2\sigma^2) + \log\sigma$
        Update θ via gradient descent
      end for
    until early stopping (patience = 10)
  end for
  return $\{f_\theta^{(b)}\}_{b=1}^B$

F Threat Model Boundary: Distillation

An adversary who generates a large synthetic dataset from the stolen model and trains a new model from scratch effectively performs *knowledge distillation*. Because the student is initialised independently and never sees the owner's training data directly, per-example memorisation of $\mathcal{W}$ is not expected to transfer.

Experiment. We generate 50,000 synthetic CIFAR-10 images from Model A via 10-step DDIM sampling and train a student model from scratch (100,000 iterations, batch size 128, fresh random initialisation). Table 17 reports t-error statistics on $\mathcal{W}$.

The student's mean t-error on $\mathcal{W}$ is 106.9, compared to 28.7 for the owner—a $3.72\times$ gap. The verification protocol correctly returns FAIL: the consistency criterion (C1) rejects because the student's score distribution differs significantly from the owner's ($p \approx 0$), indicating the student does not reproduce the owner's per-example memorisation pattern.

Table 17: Distillation boundary experiment on CIFAR-10. The student model, trained on synthetic data from the owner, exhibits 3.72× higher t-error than the owner and fails the consistency criterion.

Model	Mean t-error	Std	Ratio vs. Owner
Model A (owner)	28.71	10.85	1.0×
Student (distilled)	106.90	36.14	3.72×
ddpm-cifar10 (reference)	703.84	38.50	24.5×
Random UNet (lower bound)	3037.31	65.75	105.8×

This outcome is expected and represents a principled boundary of the method rather than a failure mode. Distillation with fresh initialisation approximates full retraining: generating a sufficiently large synthetic dataset from a diffusion model and training a new model from scratch approaches the computational cost of the original training procedure, undermining the economic motivation for model theft. Critically, the protocol does not produce a false positive—it correctly reports that ownership cannot be established, which is the far more important guarantee.

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